HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 19 Name: _____ Marks: / 60

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

   (a) Find the equation of L.

   (b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

      (i) Find the coordinates of P.

      (ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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2. When the polynomial $g(x)$ is divided by x^2+2x+a, the quotient and the remainder are $x+13$ and $b(x-3)$ respectively, where a and b are constants. It is given that $g(x)$ is divisible by $x-3$.

(a) Find a.

(b) Is there only one possible value of b such that at least two roots of the equation $g(x)=0$ are equal? Explain your answer.

(8 marks) (DSE-2026 PI Q14)

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3. It is given that P, Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C.

(b) The tangent to C at P and the tangent to C at R intersect at the point H. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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4. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

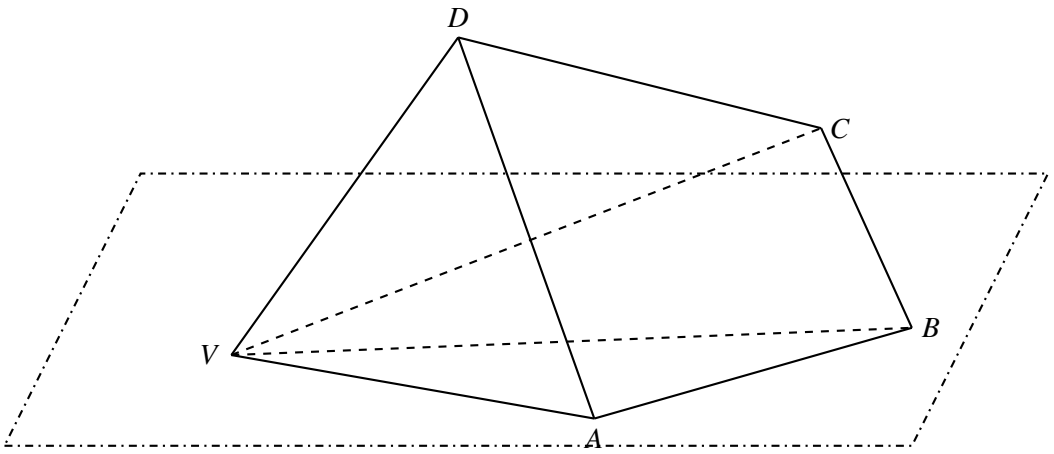


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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5. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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6. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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7. The x-coordinate of the centre of the circle C is 5. If C passes through the points $(5,0)$ and $(11,4)$, then the circumference of C is

A. 13π.

B. 14π.

C. 22π.

D. 42π.

(DSE-2026 PII Q27)

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8. A bag contains five cards numbered 2, 2, 3, 4 and 6 respectively. Two cards are randomly drawn from the bag at the same time. Find the probability that the sum of the numbers drawn is greater than 7.

A. $\frac{1}{2}$

B. $\frac{1}{5}$

C. $\frac{2}{5}$

D. $\frac{3}{10}$

(DSE-2026 PII Q28)

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9. It is given that $\log y$ is a linear function of x. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 2 and 1 respectively. If $y = ab^x$, find a.

A. 0.01

B. 0.1

C. 10

D. 100

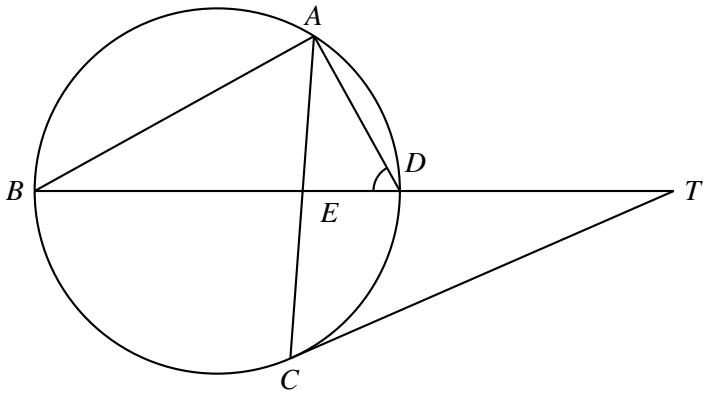
(DSE-2026 PII Q36)

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10. In the figure, $ABCD$ is a circle. AC and BD intersect at the point E. BD is produced to the point T such that TC is tangent to the circle. It is given that $AB//TC$. If $\angle AED = 85°$ and $\angle CTE = 23°$, then $\angle ADE =$



A. 39°.

B. 56°.

C. 62°.

D. 72°.

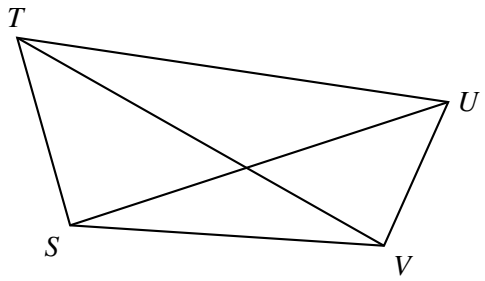
(DSE-2026 PII Q38)

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11. The figure shows the quadrilateral $STUV$, where $ST = 8$ cm and $SV = 5$ cm. If $\angle UTV = 30°$ and $\angle TSU = \angle TVU = 90°$, find $\angle SUV$ correct to the nearest degree.



A. 22°

B. 28°

C. 30°

D. 38°

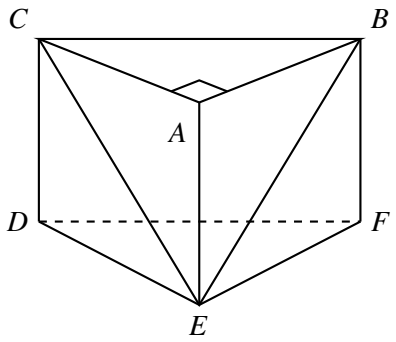
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12. The figure shows the right prism $ABCDEF$, where $AB : AC : AE = 1 : 1 : 2$. Let θ be the angle between the plane ABC and the plane BCE. If $\angle BAC = 90°$, find $\sin\theta$.



A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{\sqrt{2}}{4}$

D. $\frac{2\sqrt{2}}{3}$

(DSE-2026 PII Q40)

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13. There are 5 full-time employees and 3 part-time employees in a company. From the company, 7 employees are selected to form a queue. If no part-time employees are next to each other, how many different queues can be formed?

A. 5040

B. 14400

C. 18000

D. 32400

(DSE-2026 PII Q43)

I can finish _____ questions

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